

Wireless Magnetic Sensor Node for Vehicle Detection With Optical Wake-Up

E. Sifuentes, O. Casas, and R. Pallas-Areny

Abstract—Vehicle detectors provide essential information about parking occupancy and traffic flow. To cover large areas that lack a suitable electrical infrastructure, wired sensors networks are impractical because of their high deployment and maintenance costs. Wireless sensor networks (WSNs) with autonomous sensor nodes can be more economical. Vehicle detectors intended for a WSN should be small, sturdy, low power, cost-effective, and easy to install and maintain. Currently available vehicle detectors based on inductive loops, ultrasound, infrared, or magnetic sensors do not fulfill the requirements above, which has led to the search for alternative solutions. This paper presents a vehicle detector which includes a magnetic and an optical sensor and is intended as sensor node for use with a WSN. Magnetic sensors based on magnetoresistors are very sensitive and can detect the magnetic anomaly in the Earth's magnetic field that results from the presence of a car, but their continuous operation would drain more than 1.5 mA at 3 V, hence limiting the autonomy of a battery-supplied sensor node. Passive, low-power optical sensors can detect the shadow cast by car that covers them, but are prone to false detections. The use of optical triggering to wake-up a magnetic sensor, combined with power-efficient event-based software, yields a simple, compact, reliable, low-power sensor node for vehicle detection whose quiescent current drain is 5.5 μ A. This approach of using a low-power sensor to trigger a second more specific sensor can be applied to other autonomous sensor nodes.

Index Terms—Direct sensor interface circuit, event-based sensor wake-up, magnetoresistive sensors, sensor electronic interface, vehicle presence detection, wireless sensor node.

I. INTRODUCTION

VEHICLE detection is one of the key enabling technologies in Intelligent Traffic Systems (ITS). ITS collect and process traffic data (vehicle presence, vehicle speed, vehicle

density and occupancy ratios, etc.) from vehicle detector sensors. These can be intrusive (in-roadway) or nonintrusive (over-roadway) [1], [2]. Intrusive detectors are embedded in the pavement, taped to the surface of the roadway, or mounted on the roadway surface. This group includes inductive loops, pneumatic road tubes, piezoelectric cables, and capacitive sensors [2]. Nonintrusive detectors are placed above the surface of the roadway or on poles adjacent to it, so that they do not disrupt traffic flow during installation and maintenance. Some examples are video cameras, acoustic signal processors, radar, and ultrasonic and infrared sensors [2]. All these solutions, however, are power-hungry and expensive to deploy and maintain, hence inadequate for large-scale deployment in wireless sensor networks (WSNs), which are preferred over wired sensors networks to cover large areas.

Magnetic sensors based on magnetoresistors have recently been proposed for vehicle detection [3], [4] because they are quite sensitive, small and more immune to environmental factors such as rain, wind, snow or fog than sensing systems based on video cameras, ultrasound or infrared radiation. WSNs based on magnetoresistors can detect and track moving vehicles to obtain vehicle count or speed statistics [5], [6], but no experimental analyses are known about the use of magnetoresistors to detect static vehicles in spite of its interest, for example, to know where empty parking spots are and for street parking control. Further, published solutions rely on voltage-amplitude-based interface circuits, which are more complex, expensive and power-hungry than a direct connection of magnetoresistive sensors (AMR and GMR) to a microcontroller (MCU) [7]. Finally, there is a tradeoff between sensitivity and specificity that may result in the detection of vehicles in adjacent road lanes [8] or parking places.

This paper presents the design and implementation of a wireless sensor node to detect the presence of a vehicle in a predetermined zone. It can be useful, for example, to monitor empty places in parking lots or streets, parking meters, automatic door/gate opening, semaphore control, traffic control, and railroad crossing control. The sensor node includes a magnetic and an optical sensor. The optical sensor, that drains a very small current, is always on and detects the reduced illumination resulting from the presence of a car or any other object. The magnetic sensor is wakened up by the optical sensor and detects the presence of ferromagnetic material near the sensor. Both sensors are directly connected to a microcontroller without any intervening amplifier, which results in a simple, compact and very low-power vehicle detector. This use of a low-power sensor to wake-up a second that has a better performance can be extended to other applications.

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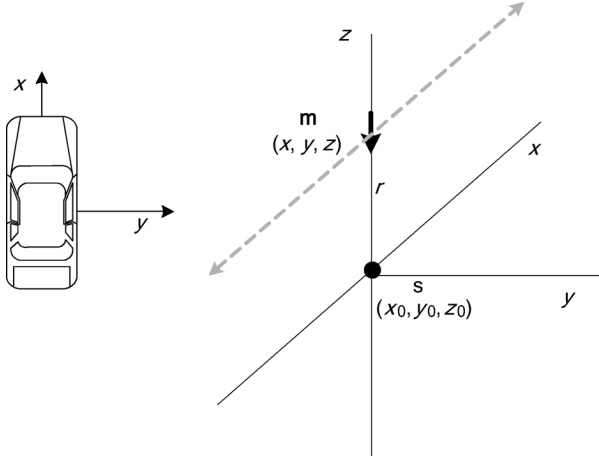


Fig. 1. Vehicle modeled as a magnetic dipole parallel to the vertical Earth's magnetic field component Z_E .

II. SENSING APPROACH

A. Magnetic Sensor

All road vehicles have significant amounts of ferrous metals in their structure and parts, and produce a magnetic anomaly because their magnetic permeability is much higher than that of the surrounding air and soil. This concentrates and/or distorts the flux lines of the Earth's magnetic field and increases the magnitude of the magnetic flux density (or magnetic induction) $\mathbf{B}$ inside and near the vehicle. A magnetic sensor close or below a vehicle will detect this change in the Earth's magnetic field.

The Earth's magnetic field is uniform over wide areas, but its vector direction changes from horizontal at the Equator to almost vertical at the magnetic poles. In Barcelona, Spain, the current vertical component, $Z_E = 38 \mu\text{T}$, is larger than the horizontal component, $H_E = 25 \mu\text{T}$, the strength of the field is $F_E \approx 45 \mu\text{T}$ and the field vector points 60° down from horizontal towards the magnetic north pole. Therefore, a magnetic z axis sensor can sense the vertical component and detect its change when a vehicle passes or is parked over the sensor.

The simplest mathematical model to describe the magnetic signature of a vehicle is a magnetic point dipole [8], [9] with a magnetic moment $\mathbf{m}$ centered in the vehicle and parallel to the Earth's field (Fig. 1). The field components B_x , B_y , B_z produced by $\mathbf{m}$ are [10]

$$B_x = \frac{\mu_0}{4\pi} \frac{m_x(2x^2 - y^2 - z^2) + 3m_yxy + 3m_zxz}{r^5} \quad (1a)$$

$$B_y = \frac{\mu_0}{4\pi} \frac{3m_xxy + m_y(2y^2 - x^2 - z^2) + 3m_zyz}{r^5} \quad (1b)$$

$$B_z = \frac{\mu_0}{4\pi} \frac{3m_xxz + 3m_yyz + m_z(2z^2 - x^2 - y^2)}{r^5} \quad (1c)$$

where μ_0 is the permeability of free space, m_x , m_y , and m_z are the magnetic dipole moments in x -, y -, and z -direction, respectively, and r is the distance from the dipole to the observation point.

Fig. 1 shows a magnetic dipole moment placed in (x, y, z) and an observation point s in (x_0, y_0, z_0) . We assume that the

magnetization induced by the Earth's field (assumed itself a homogeneous magnetic field in z direction) is parallel to the longitudinal axis of the dipole and perpendicular to the longitudinal axis of the vehicle. If the vehicle moves along the x axis, with $y = 0$, and we assume $m_x = m_y = 0$, that is to say only m_z is considered, the magnetic induction is

$$B_m = Z_E + B_Z = Z_E + \frac{\mu_0}{4\pi} \frac{m_z(2(z - z_0)^2 - (x - x_0)^2)}{[(z - z_0)^2 + (x - x_0)^2]^{\frac{5}{2}}} \quad (2)$$

where Z_E is the vertical component of the Earth's magnetic field and B_Z is the vertical magnetic anomaly created by the vehicle (magnetic dipole).

Vehicles apart from a magnetometer may look like a magnetic point dipole but as a vehicle consists of several ferrous metal components (e.g., engine and wheel axis), the magnetic dipoles of these individual components will reveal when the sensor is close enough. Therefore, we can expect each vehicle to yield a distinct magnetic signature with multiple peaks corresponding to its multiple dipoles. In practice, a given magnetic anomaly can be produced by a small vehicle close to the sensor or by a larger vehicle farther away, maybe in the next road lane or parking row.

B. Optical Sensor

The illumination level outdoors (i.e., in roads, streets and parking areas) depends on the natural and artificial light sources available. For example, the illuminance on a horizontal surface (at floor level) can be 100 klx on a clear day, 5 klx on a cloudy day and below 100 lx at night when there is a single street lamp [11]. Vehicles or any other object can reduce the illuminance underneath them to less than 50 lx (measured at night, clear and cloudy days for different cars using an ISO-Tech Lux-1335 light meter, whose resolution is 0.01 lx [9]).

Therefore, a light meter can detect an object that covers the sensor. To assess whether that object is actually a car, the optical sensor can wake-up a magnetic sensor, which will distinguish ferromagnetic objects from any other materials. Hence, a passive, low-cost optical sensor such as an LDR, able to run with very little power, combined with a more selective (and power demanding) magnetic sensor can result in a reliable, low-cost and low-power vehicle detector system suitable for wireless sensor nodes.

III. WIRELESS SENSOR NODE DESIGN AND IMPLEMENTATION

A. Sensor Node Architecture

The main components of a wireless sensor node are: a sensor, a processor, a transceiver, and an energy source. Fig. 2 shows a prototype of the wireless sensor node proposed. The processor is a low-cost, low-power microcontroller (MSP430F2274) (running at 8 MHz, quartz oscillator clock) that has five low-power modes, a wake-up time shorter than 6 μs , and a versatile and friendly programmable clock system. All the eight bits of its I/O port P2 have edge-selectable interrupt input capability. The magnetic sensor is the AAH002 (NVE), a compact and low-cost GMR (Giant Magnetoresistive) sensor with two sensing resistors connected in opposed arms of a Wheatstone bridge, $R_0 =$

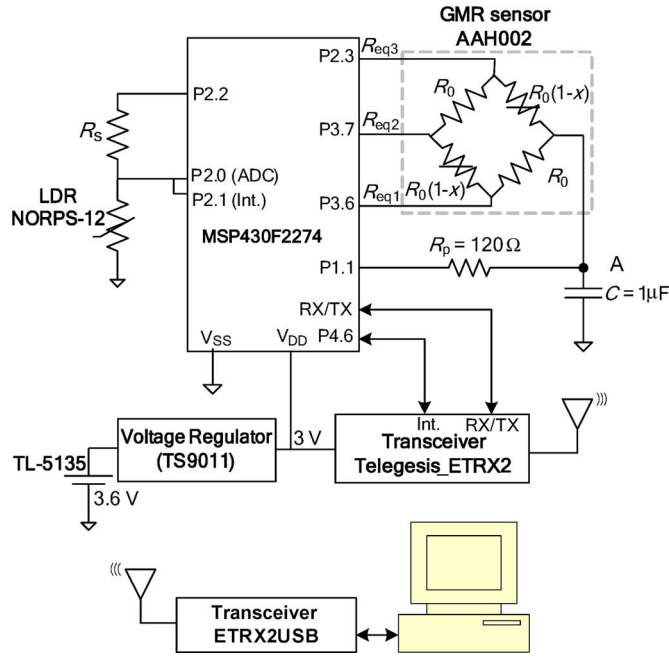


Fig. 2. Wireless magnetic sensor node for vehicle detection based on a wake-up system triggered by a high-resistance light sensor.

2 kΩ (±20%), a typical sensitivity of 11 mV/V/G, and temperature range from −50 °C to 150 °C. The optical sensor is an LDR NORPS-12 (CdS, SILONEX) with dark resistance 1 MΩ (1 lx), light resistance 5.4 kΩ (1000 lx) and temperature range from −50 °C to 150 °C.

The transceiver (Telegesis ETRX2) is a low-power, 2.4 GHz free-licensed ISM band module based on the EM250 (Ember) single chip Zigbee/IEEE802.15.4 solution, and intended for networking. The configurable functionality often allows the ETRX2 wireless meshing module to be used without an additional host MCU, thus saving more time and costs. All circuits were supplied from a 3 V voltage regulator (TS9011) connected to a lithium primary battery (TL-5135, Tadiran) that can supply 20 μA at 3.6 V for ten years, and can operate from −30 °C to 70 °C.

B. Vehicle Detection Algorithm

Magnetic induction has been measured using the method described in [7] for resistive sensor bridges, which is based on measuring the discharge times of C (Fig. 3) through selected resistive paths that include the sensing bridge arms. Because the AAH002 GMR sensor implements a half-bridge topology, B can be obtained from

$$B = \frac{x^*}{(x^* - 2)S} \quad (3)$$

where S is the sensitivity of the GMR sensor and x^* is the estimated fractional resistance change of the sensors in the bridge, calculated from the corresponding time-based equation in [7]. When the LDR sensor detects a change in illuminance that is large enough for the voltage in P2.1 to trigger an interrupt request, the MCU is waken up, measures B , calculates the change

with respect to a stored baseline value (the vertical component of the Earth's magnetic field in the absence of any car) and checks whether this change exceeds or not a given threshold. A positive result means that a car covers the sensor. Measurements for different car models using a commercial magnetometer with 0.1 nT resolution (Mag-01H, Bartington), suggest a 5 μT threshold for car detection [9]. The baseline magnetic field can be determined from a site survey before installing the sensors.

Fig. 3 shows the detection algorithm in detail. Table I shows the activities and operation modes of the sensor node. The default state for the node is sleep mode (MCU in LPM4 mode and ETRX2 in mode 3, their lowest power consumption modes, activity 13 in Table I) and the average current consumption is 5.5 μA. A reduction in illuminance below 50 lx brings the voltage in ports P2.0 (configured as ADC input) and P2.1 (configured as external interrupt pin) above the logic voltage threshold needed to request an external interrupt and wakes up the sensor node, starts up the external quartz oscillator clock, and waits for a warm up time of 3 ms (activity 2). Then, B is measured and the change with respect to its previously stored baseline value is calculated and compared with the preset threshold (activities 3–6). If the comparison is positive, it is assumed that there is a vehicle, the transceiver is placed in mode 0 and sends a message to the base station (activities 7, 11, and 12).

The message contains the sensor node ID, the actual measured value B , and ends with an “1” to indicate that the parking spot is occupied. Afterwards, the sensor node changes from interrupt detection mode to periodic detection (polling mode) by running the WDT (Watchdog timer) at 1.5 kHz and measuring B each 20 s to detect whether the vehicle is still parked. During this 20 s interval, the sensor node is placed in low-power mode (MCU in LPM4a mode and ETRX2 in mode 3, activity 10) waiting for a WDT interrupt, so the average current consumption is only 2.6 μA. When the vehicle leaves the detection zone, the transceiver sends a message to the base station that ends with a “0,” to indicate that the place is empty.

If, when in “dark condition,” the change in B is less than 5 μT, meaning that no vehicle is detected, no message is sent. The illumination level E is then measured (activity 8) by the 10 bit ADC embedded in the MCU which is connected to the voltage divider formed by the LDR sensor and R_s (port P2.0 in Fig. 2). If E is less than 50 lx and the change in B is less than 5 μT, we either have: (a) a nonferromagnetic object covering the sensor; (b) a low-level local illumination (e.g., it is night time or the sensor window is dirty); or (c) a vehicle parked nearby that projects a shadow onto the sensor node. Whatever the situation, to be able to detect a vehicle arriving at the sensor, B and E are measured every 20 s, until E exceeds 50 lx.

If E rises above 50 lx and the change in B is less than 5 μT, the sensor node reconfigures P2.1 (IES) and returns to sleep mode (activity 13) to wait for the next interrupt event (reduced illumination), which would indicate that another possible vehicle has arrived. If E stays below 50 lx for some time and B does not increase above 5 μT, it may be decided that the sensor node needs some maintenance. Information from nearby sensors can also be considered in taking that decision.

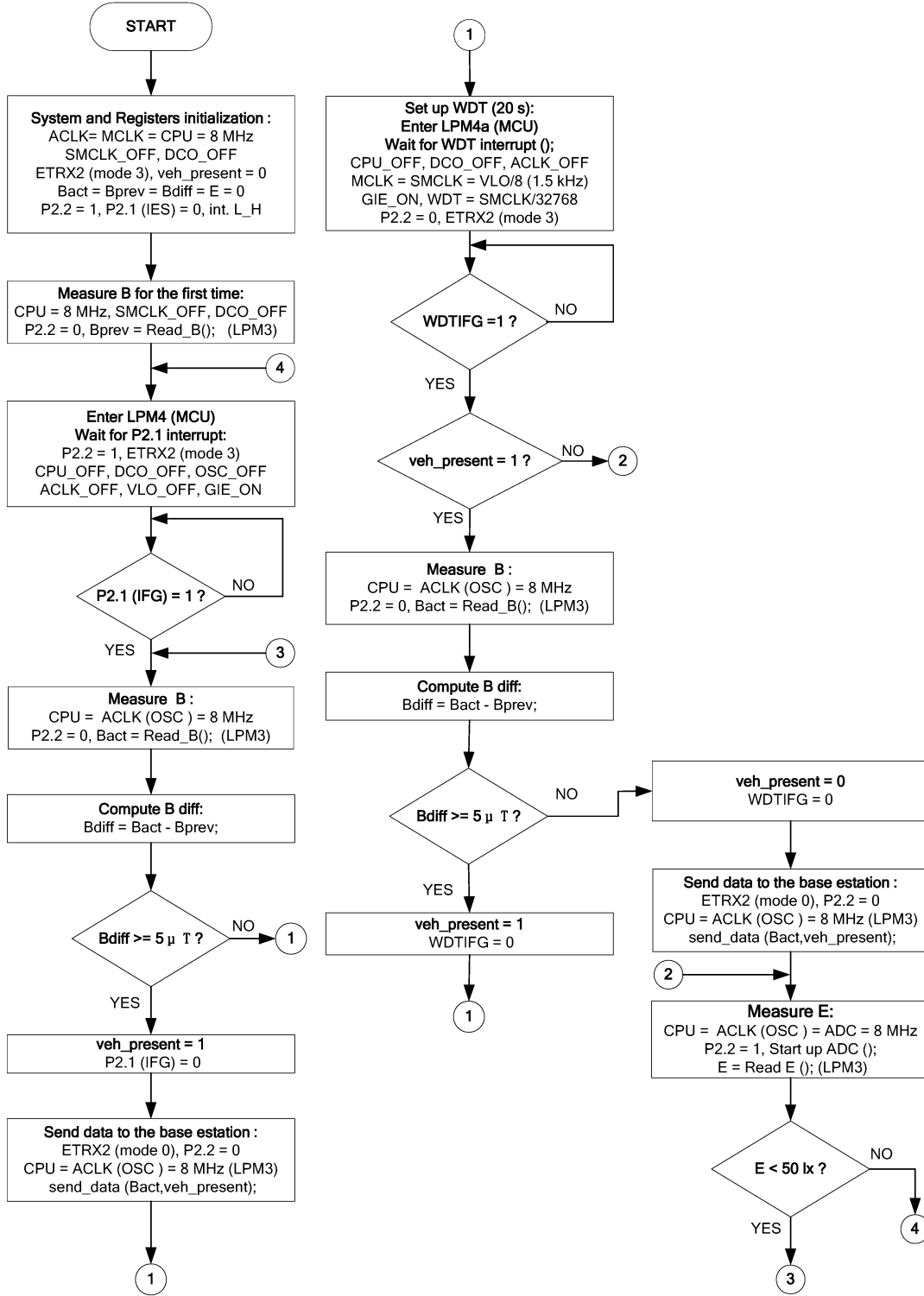


Fig. 3. Event-based algorithm to detect vehicle presence or absence.

C. Power Consumption Analysis

To calculate the average power consumption of a sensor node, we need to know the power consumption and duration of each activity performed. The average power consumption is

$$P_m = I_m V_{DD}|_{3.0 \text{ V}}$$

where V_{DD} is the supply voltage and I_m is the average total current consumption,

$$I_m = \sum_{i=0}^n \left(\frac{t_i}{T} I_i \right) + \left[1 - \sum_{i=0}^n \left(\frac{t_i}{T} I_i \right) \right] I_s \quad (5)$$

where i identifies an active sequence, I_i is the average current consumption of sequence i , t_i is the duration of i , I_s is the cur-

TABLE I
TIME DURATION AND CURRENT CONSUMPTION FOR EACH ACTIVITY OF THE SENSOR NODE. R_{EQ1} , R_{EQ2} , AND R_{EQ3} ,
ARE THE EQUIVALENT RESISTANCES OF THE THREE DIFFERENT DISCHARGING PATHS FOR C DEPENDING ON THE
VOLTAGE/IMPEDANCE SELECTED AT PINS 2.3, 3.7, AND 3.6 OF THE MCU [7]

Activities	MCU	ETRX2	t/ms	$I/μA$
1. Start up, initialize variables and registers.	AM1	Mode 3	0.45	0.8
2. External oscillator warm up (8 MHz), voltage divider disabled (P2.2 = 0).	AM2	Mode 3	3.00	1.2
3. Measure B , voltage divider disabled (P2.2 = 0).	-	-	-	-
3.1. Charge C across R_p .	LPM3	Mode 3	0.83	5.8
3.2. Discharge C and time measurement across R_{eq1} .	LPM3	Mode 3	1.61	0.4
3.3. Charge C across R_p .	LPM3	Mode 3	0.83	5.8
3.4. Discharge C and time measurement across R_{eq2} .	LPM3	Mode 3	2.24	0.4
3.5. Charge C across R_p .	LPM3	Mode 3	0.83	5.8
3.6. Discharge C and time measurement across R_{eq3} .	LPM3	Mode 3	1.60	0.4
3.7. Compute B .	AM3	Mode 3	0.79	2.8
4. Compute B change ($B_{act} - B_{prev}$).	AM3	Mode 3	0.04	2.8
5. Port initialization to charge C across R_p .	AM3	Mode 3	0.01	2.8
6. Port initialization to discharge C across R_{eqi} .	AM3	Mode 3	0.01	2.8
7. UART Configuration.	AM3	Mode 3	0.02	2.8
8. Measure E .	-	-	-	-
8.1. Start up ADC at 8 MHz and AD conversion, voltage divider enabled (P2.2 = 1).	LPM3	Mode 3	0.03	2.8
8.2. Compute E , voltage divider disabled (P2.2 = 0).	AM3	Mode 3	0.80	2.8
9. Configure WDT (Watchdog Timer) for 20 s, enter sleep mode MCU.	AM3	Mode 3	0.06	2.8
10. Delay 20 s, wait for WDTIFG interrupt, voltage divider disabled (P2.2 = 0).	LPM4a	Mode 3	20×10^3	2.6×10^{-3}
11. Start up RF module, voltage divider disabled (P2.2 = 0).	-	-	-	-
11.1. Wakeup RF module.	AM3	Mode 0	5.00	11.5
11.2. Send data.	AM3	Mode 0	13.54	39.4
11.3. Listen for reply.	LPM3	Mode 0	50.00	8.7
12. Turn off UART, enter sleep mode RF.	AM3	Mode 3	0.02	11.5
13. Asleep, voltage divider enabled (P2.2 = 1).	LPM4	Mode 3	-	5.5×10^{-3}

AM1: SMCLK = VLO = ACLK = Off, MCLK = CPU = DCO = 1 MHz.

AM2: SMCLK = VLO = Off, MCLK = CPU = DCO = 1 MHz, ACLK = 8 MHz. (Note A).

AM3: SMCLK = VLO = DCO = Off, MCLK = CPU = ACLK = 8 MHz. (Note A).

LPM3: SMCLK = VLO = DCO = CPU = Off, MCLK = ACLK = 8 MHz. (Notes A and B).

LPM4: SMCLK = VLO = DCO = MCLK = CPU = ACLK = Off (GIE_On), (Note A)

LPM4a: DCO = CPU = ACLK = Off (GIE_On), SMCLK = MCLK = VLO/8 (1.5 kHz), WDT enabled, (Note A).

Mode 3: MCU = RADIO = Asleep, TIMERS = Off.

Mode 0: MCU = RADIO = Awake, TIMERS = User defined.

Note A: All internal peripherals are disabled, and non-used ports are configured as digital output in Low state.

Note B: Only Timer A and capture module are enabled, except in activity 8 where ADC is enabled.

rent consumption in sleep mode, and T is the clock period. From I_i , t_i , I_S and the duty cycle, we can calculate I_m .

Table I shows the measured current consumption and duration of each activity of the sensor node, and the operation modes of the MCU and transceiver (ETRX2). Table II shows that the average current consumption of the sensor node in sleep mode (no vehicle detected, no data transmitted) is $5.5 \mu A$, so the battery (1700 mAh) would last about 35 years, hence longer than the battery lifetime specified by the manufacturer (ten years). The average current consumption in active mode largely depends on the sensor application. For example, for automatic door/gate opening, the worst case could be a vehicle detection every 20 s, hence 4320 detections and 8640 data packets sent each day,

TABLE II
CURRENT CONSUMPTION OF THE SENSOR NODE IN SLEEP MODE

Components	Mode	$I_s/\mu A$
MCU	LPM4	0.1
MCU	LPM4a	0.5
ETRX2	3	0.8
TS9011	Active	1.3
LDR sensor and $R_s = 910 \text{ k}\Omega$	Active	3.3
Sensor node	Sleep	5.5

whereas common cases could be 500 detections a day for a community and 10 vehicles a day for a single-family use (automatic garage door). Table III shows the corresponding average current

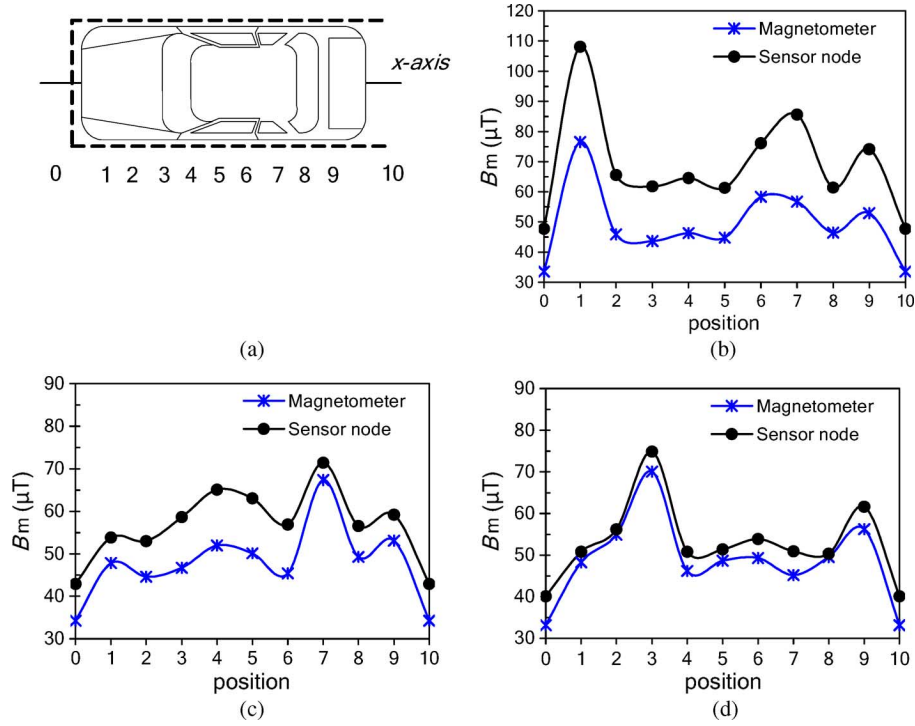


Fig. 4. Magnetic signature of three different cars. (a) Measurement points were along a centered line below each parked car. (b) Ford Focus. (c) Opel Meriva. (d) Fiat Punto.

TABLE III
AVERAGE CURRENT CONSUMPTION OF THE SENSOR NODE IN
THREE DIFFERENT SCENARIOS

Vehicles	Sensor node	t/s	I_m/mA	Lifetime /years
4320	Active Sleep	1615 84785	0.130	1.5
500	Active Sleep	187 86213	0.020	9.7
10	Active Sleep	3.73 86396.27	5.78×10^{-3}	33.5

consumption of the sensor node in active mode for these three cases. The lifetime of the sensor node is greatly extended by using software and hardware based on events, such as the optical triggering wakeup system proposed.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

To test the sensor node proposed, we built a minimal WSN with the sensor node in Fig. 2 acting as a Sleep End Device (SED) and an ETRX2USB (Telegesis) module acting as a network coordinator node connected to a laptop computer by an USB interface. We measured on parked vehicles and during parking maneuvers in an outdoor parking area. Two compact vehicles (Ford Focus and Opel Meriva) and a subcompact (Fiat Punto) were involved. Tests were repeated at different times of the day, hence different natural illumination levels and night lights. The optical sensor detected the presence of a car in all instances, including car arrival and departure, and woke up the sensor node, which send the corresponding data packet to the coordinator node.

Fig. 4 shows the magnetic induction simultaneously measured by the sensor and the Mag-01H magnetometer for the three vehicles when parked. Measurements were performed at floor level and ten different positions along a line centered in the vehicle [Fig. 4(a)]. For the sensor node, each B_m value was estimated from (3) by averaging ten measurements of x^* [7], [9]. The resolution achieved was 750 nT (8 bit) for a measurement range of 200 μT , and the measurement time was 112 ms. Averaging 100 values increased the resolution to 10 bit (187 nT) but the measurement time extended to 1.12 s. Fig. 4(b)–(d) show the respective magnetic signatures for each car model. These signatures suggest that compact cars can be roughly described by three magnetic dipoles: two larger dipoles for the engine and wheel axes and a smaller dipole between them. The sensor always measured larger B_m values than the magnetometer, perhaps because of the large tolerance for the magnetoresistors ($\pm 20\%$), but that difference did not affect car detection.

Fig. 5 shows the corrected magnetic induction values ($B_z = B_m - Z_E$) obtained during parking maneuvers. The sensor node was placed at the center of the parking spot. B was measured each 120 ms when the vehicle arrived, stayed parked for about 10 s and left. The magnetic signature consists of a first peak that corresponds to the engine and front wheel axis passing over the sensor when the vehicle arrives, followed by a constant B_z when the car is parked, and a second peak that corresponds to the engine and front wheel axis when the car backs to leave the parking place. Results when the vehicles backed to park yielded similar curves.

Vehicles parked at spots about 1.5 m from the sensor node produced negligible changes in the B_m baseline value. This is probably because of the high directivity of the magnetic sensor and raises the question of the actual interest of using an additional (optic) sensor. Nevertheless, a continuous measurement

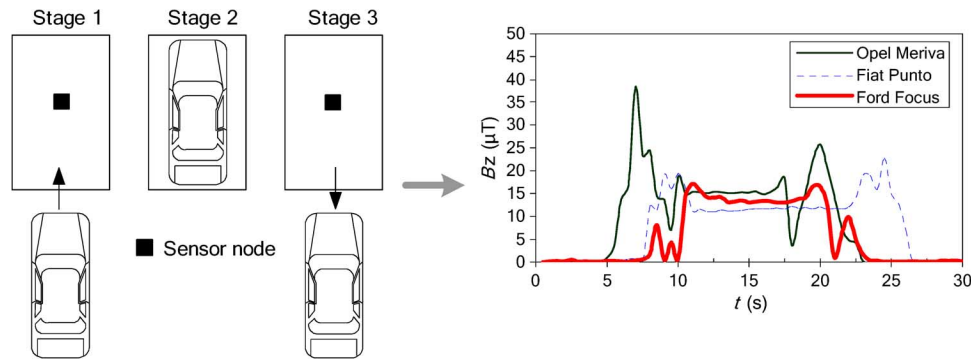


Fig. 5. Magnetic vehicle signatures for three different cars parking over the sensor.

of the magnetic field using a 2 k Ω sensor bridge supplied at 3 V would drain about 1.5 mA, day or night. A voltage divider with an LDR sensor drains far less current; the power drain of the sensor node proposed was about 5.5 μ A when in sleep mode, 60% of which were from the voltage divider that included the light sensor and $R_s = 910$ k Ω (Fig. 3). Power consumption will depend on how often the magnetic sensor is asked to measure, and on the use of the transceiver.

To avoid unnecessary calls to duty, for example by shadows cast by nearby cars, particularly at night, the actual illumination level could be measured to update an internal threshold value and use it for future detections. This could also compensate for possible changes in the transparency of the sensor housing, for example because of dirt.

V. CONCLUSION

We have proposed a wireless sensor node based on two passive sensors to detect the presence of a vehicle in a pre-determined zone. When a vehicle arrives or departs, an LDR sensor detects the change in illumination and wakes up the sensor node. Then, a magnetoresistive sensor measures the magnetic anomaly in the Earth's magnetic field created by the presence of a vehicle, if any, and the necessary actions are taken (data transmission, control signals issued). Otherwise, the sensor node falls back in sleep mode, where it drains a very small current (5.5 μ A in our prototype). To further reduce power consumption, both sensors are directly connected to the ports of a microcontroller without any analog front end. The measurement time exceeds 100 ms, which would be too long to detect fast vehicles, but that is short enough for a simple, compact, low-cost, and very low-power vehicle detector for parking control, and that is also reliable, sturdy and easy to install and maintain. Amplitude-based sensor interfaces would provide faster measurements at the cost of a reduced battery life.

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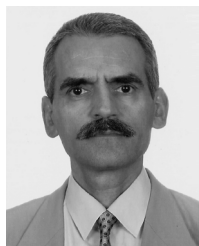
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